Combine Harvester



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paddy combiner (small)

What it is used for

A small paddy combine harvester cuts rice plants and combines harvesting operations such as threshing and grain separation in one machine. Correct field preparation, machine adjustment, and safe operation help reduce grain loss, crop damage, and machinery accidents.

Main parts

- Cutting header and crop divider
- Reel or crop-feeding mechanism, if fitted
- Threshing drum and concave
- Straw walkers or grain separation system
- Grain cleaning fan, sieves, and collection tank
- Engine, drive system, controls, guards, and discharge outlet

Before you start

1. Read the operator manual and confirm the machine is suitable for the paddy crop, field conditions, and expected crop height.
2. Inspect the engine, fuel system, belts, chains, cutting blades, threshing parts, tyres or tracks, guards, and controls.
3. Check engine oil, coolant, fuel, lubrication points, and other fluid levels according to the manual.
4. Remove people, animals, stones, wire, and other obstacles from the field and harvesting route.
5. Check soil firmness, water depth, slopes, field access, and turning space; avoid unsafe or excessively soft ground.
6. Adjust cutting height, threshing clearance, fan speed, and sieve settings according to crop moisture and machine instructions.

How to use

1. Park the harvester on firm, level ground and complete the pre-start inspection.
2. Ensure guards are fitted, the discharge area is clear, and all bystanders are at a safe distance.
3. Start the engine using the manufacturer's procedure and check for warning indicators, leaks, or abnormal sounds.
4. Allow the engine and relevant systems to reach the recommended operating condition before harvesting.
5. Engage the cutting and threshing systems using the prescribed sequence and confirm they operate normally.
6. Move into the crop at a controlled speed and keep the header aligned with the standing paddy.

7. Maintain a steady forward speed so the crop feeds evenly without overloading the threshing system.
8. Monitor grain flow, grain cleanliness, straw discharge, engine load, and visible crop losses.
9. Adjust travel speed or approved machine settings if grain loss, unthreshed ears, or excessive broken grain appears.
10. Stop on firm ground before clearing blockages; disengage the drives, switch off the engine, remove the key, and wait for all moving parts to stop.
11. Clear crop material only with approved tools and according to the manual; never reach into moving mechanisms.
12. Unload collected grain at a designated safe location, keeping people clear of the discharge mechanism.
13. At the end of harvesting, disengage the systems, park securely, switch off the engine, and allow hot components to cool.
14. Clean crop residue from accessible areas and inspect the machine for damage before storage or transport.

⚠ Safety

- Only trained operators should operate the harvester; follow the manufacturer's instructions and local agricultural safety requirements.
- Keep all guards and shields fitted and never approach the cutting header, belts, chains, or threshing drum while they are moving.
- Never clear blockages with the engine running or stored mechanical energy present.
- Keep people and animals away from the machine, especially during reversing, turning, and grain unloading.
- Do not operate on steep slopes, unstable embankments, deep water, or ground that cannot safely support the machine.
- Refuel only with the engine stopped and cooled, away from ignition sources; keep suitable fire-fighting equipment accessible.
- Wear suitable footwear, eye protection, and hearing protection as required; keep loose clothing away from moving parts.

Care & storage

- Remove straw, dust, and chaff after shutdown, paying attention to hot engine and exhaust areas only after they have cooled.
- Inspect blades, belts, chains, bearings, sieves, and threshing components for wear or damage.
- Check engine oil, coolant, fuel filters, and lubrication points at the intervals specified in the manual.
- Clean air filters and inspect the grain cleaning system regularly.
- Store the harvester in a dry, secure area and arrange scheduled servicing before the next season.

Common mistakes to avoid

- Operating at excessive speed for the crop density or field condition.
- Using unsuitable cutting height, threshing clearance, or cleaning settings.

- Clearing a blockage before fully stopping and isolating the machine.
- Ignoring grain loss, abnormal vibration, overheating, or damaged guards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.